

DAN KLUSKIEWICZ

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Summary

Applied researcher with a geophysics background, building Python packages and statistical models for large remote-sensing and natural-hazard problems: individual-tree forest inventory from imagery, LiDAR, and field data, and probabilistic models of flood hazard and wildfire.

Skills

Software: Python package development, reproducible workflows, large-scale data processing, model training and validation

Geospatial data: aerial imagery, LiDAR, field observations, spatial sampling

Modeling: object detection, classification, regression, and uncertainty estimation

Work Experience

Foresite — Moscow, ID (working in Index, WA)

September 2018 – Present

Programmer Statistician

Lead research and development for tools that predict individual-tree forest inventory from field data, aerial imagery, and LiDAR, and that report validation statistics and uncertainty with the inventory.

- Developed a reproducible end-to-end workflow that organizes multi-terabyte remote-sensing data, incorporates field data to train models that detect and classify trees, evaluates models with validation statistics, and scales inference across large areas.
- Built Python packages to support and implement this workflow, and to estimate and document uncertainty in our inventories.
- Worked with researchers, engineers, foresters, and biometricians to maintain this workflow and to design field-sampling protocols that balance cost, logistics, and statistical precision.
- Presented inventory products and validation results to clients and at conferences (domestic and international).

Talus Analytics — Boulder, CO

June 2016 – April 2018

Quantitative Researcher

- Proposed a geometric method for probabilistic flood-hazard assessment and led its implementation in Python.
 - Implemented a simplified static wildfire-spread model from scratch, then extended it to time-dependent predictions.
 - Developed a wildfire accumulation model that delineates areas by hazard and predicts probable maximum losses within them.
 - Wrote the methods section for a FEMA publication on rapid flood modeling.
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Education

University of Washington

M.S. in Geophysics

2015

GPA: 3.94

NSF Graduate Research Fellowship

Pennsylvania State University, Schreyer Honors College

B.S. in Physics and Mathematics

2010

GPA: 3.99

Highest Distinction, Class rank **1** out of **519** (College of Science)

Publications

- Kluskiewicz, Dan and others. (2017). Sonic methods for measuring crystal orientation fabric in ice, and results from the West Antarctic ice sheet (WAIS) Divide. *Journal of Glaciology*. 63. 1–15. [doi:10.1017/jog.2017.20](https://doi.org/10.1017/jog.2017.20).